English

(1001CJA102922062)

Test Pattern

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
PART TEST
18-12-2022

JEE(Main+Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS, TNAS, I(A) & LIVE-I(A)] (SCORE-I)

Time : 3 Hours **Maximum Marks : 180**

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble ◯ **COMPLETELY.**
12. The correct way of darkening a bubble is as : ●
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **g = 10 m/s²** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS	
Atomic No. :	H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58
Atomic masses :	H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca=40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba=137, Ce = 140

• **Boltzmann constant**	$k = 1.38 \times 10^{-23}\ JK^{-1}$
• **Coulomb's law constant**	$\frac{1}{4\pi\varepsilon_0} = 9\times10^9$
• **Universal gravitational constant**	$G = 6.67259 \times 10^{-11}\ N\text{–}m^2\ kg^{-2}$
• **Speed of light in vacuum**	$c = 3 \times 10^8\ ms^{-1}$
• **Stefan–Boltzmann constant**	$\sigma = 5.67 \times 10^{-8}\ Wm^{-2}\text{–}K^{-4}$
• **Wien's displacement law constant**	$b = 2.89 \times 10^{-3}\ m\text{–}K$
• **Permeability of vacuum**	$\mu_0 = 4\pi \times 10^{-7}\ NA^{-2}$
• **Permittivity of vacuum**	$\in_0 = \frac{1}{\mu_0 c^2}$
• **Planck constant**	$h = 6.63 \times 10^{-34}\ J\text{–}s$

Space for Rough Work

HAVE CONTROL ⟶ HAVE PATIENCE ⟶ HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : –2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in –2 marks.

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1. A certain microwave oven projects 1.77 kW of microwaves onto a 100 cm by 20 cm area.

(A) The intensity of microwaves is 8.85×10^3 W/m^2

(B) The peak intensity of microwaves is 8.85×10^3 W/m^2

(C) The square of peak electric field strength in the waves is 6.66×10^6 N^2/c^2

(D) The peak value of magnetic energy density 2.95×10^{-5} J/m^3

Space for Rough Work

2. Mark the correct statements about polarisation of light.

(A) If a polaroid is kept in front of you and rotated while looking at the light reflected from water or glass, you will notice that light gets bright and completely dark sequencially.

(B) At Brewster's angle of incidence the plane of vibration of electric field vector of reflected light is perpendicular to plane of reflection

(C) A polaroid consists of long chain molecules alinged in direction parallel to the axis of polaroid which absorb the electromagnetic energy of light.

(D) If θ_1 is the Brewster's angle for light reflected from the top of an interface between two substances and θ_2 is the Brewster's angle for light reflected from below, then $\theta_1 + \theta_2 = 90°$.

3. In a single-slit diffraction pattern the central fringe is 450 times as wide as the slit. The screen is 18000 times farther from the slit than the slit is wide. The wave length of light is λ and width of slit is 'W'. [$\pi^2 = 10$]

(A) The ratio $\frac{W}{\lambda}$ is 80.

(B) The distance of second minima from the point of maximum intensity is 450 W

(C) The distance of second maxima from the point of maximum intensity is 450 W

(D) If the angular position of the point where intensity is 0.9 times the maximum intensity is θ, then $\sin\theta \approx \frac{1}{480}$.

Space for Rough Work

4. Mark the correct statements of the following :

(A) According to Rayleigh's criterion for resolution, two images would be just resolved when central maxima of one image coincides with central maxima of the other.

(B) The resolving power of a microscope is inversely proportional to the focal length of the objective lens of microscope.

(C) The resolving power of a telescope is directly proportional to the aperture diameter of the objective lens.

(D) For an eagle flying at an altitude of 100 m having pupil diameter of 6 mm, two point objects at a separation of 20 mm can be resolved considering wave length of light $\lambda = 600$ nm.

5. A source of sound radiates uniformly in all directions. Along a radial line from the source locate two points separated by 3 m such that intensity at the nearer point is 4dB above that of more distance point. [given $10^{0.2} = 1.6$]

(A) Ratio of intensities at the nearer point to the distance point is 1.6

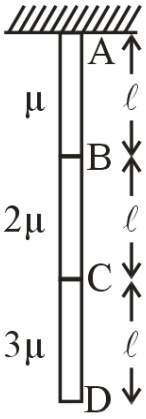
(B) Ratio of distance of distant point to the nearer point from the source is 1.6

(C) Distance of the nearer point from source is 5 m.

(D) Distance of the distant point from source is 8 m.

Space for Rough Work

6. A composite wire is made by joining 3 wires of length ℓ and linear mass density μ, 2μ and 3μ respectively as shown in figure. Wire AB has the least density followed by BC and CD.
A quick transverse pulse is given at the end D at t = 0.



(A) The ratio of time taken by pulse to travel the DC, DB and DA is $1 : \sqrt{2} : \sqrt{3}$

(B) The ratio of time taken by pulse to travel the DC, CB and BA is $1 : \frac{1}{\sqrt{2}} : \frac{1}{\sqrt{3}}$

(C) The ratio of length of the pulse just above B and just below C is $\sqrt{5} : 1$

(D) Acceleration of the pulse travelling upwards is in the ratio of 3 : 5 : 6 in the strings CD, BC and AB respectively.

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
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Negative Marks : –1 In all other cases.

Paragraph for Questions 7 and 8

A detector of radio waves in a radio astronomical observatory is placed on the sea beach at a height h = 2.10 m above the sea level. After the rise of a star, radiating electromagnetic waves of wavelength λ = 21 cm, above the horizon the detector registers series of alternating maxima and minima. The registered signal is proportional to the intensity of detected waves. The detector registers waves with electric vector vibrating parallel to the sea surface.

7. If the angle between the star and horizon in the moment the detector resistors maxima for the second time is θ_0, the value of $\sin\theta_0$ is given by $N \times 10^{-3}$. Find the value of N.

(A) 55 (B) 65 (C) 75 (D) 85

8. At reflection of the electromagnetic wave on water surface the ratio of the intensities of the electric field of the reflected and (E_r) and incident (E_i) wave follows the law :

$$\frac{E_r}{E_i} = \frac{n - \cos i}{n + \cos i}$$

where n is the refractive - index and i is the incident angle of wave. For the surface "air-water" for λ = 21 cm, the refractive index n = 9

If the signal ratio of the first maximum to the next minimum is $R \times 10^4$, find the value of R.

(A) 3.25 (B) 4.25 (C) 5.25 (D) None of these

Space for Rough Work

Paragraph for Questions 9 and 10

Three successive resonance frequencies for a certain string are 175, 245 and 315 Hz. The length of the string is 12.87 m. The separation between the nodes in the respective modes are α, β and γ in meters. the fundamental frequency of the this string is f_0 mass per unit length of the string is 10^{-3} kg/m.

9. Find the sum of α, β and γ.

(A) 6.56 (B) 7.18

(C) 6.18 (D) 11.68

10. The same string resonates with the frequency f_0, but has two nodes and two antinodes on the string. Find the tension in the string in newtons.

(A) 360.72 (B) 260.72

(C) 266.72 (D) 366.72

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;
Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

A massless speaker emitting sound at 600 Hz is placed on top of a mass connected to a spring. The mass spring system is set in an oscillatory state, vibrating horizontally at 4 Hz with amplitude A. The velocity of sound is given to be 330 m/s.

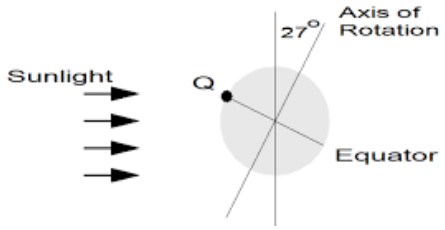
1. If the difference between highest and lowest frequencies heard by a stationary observer on the line of motion is 2 Hz, find the amplitude A in cm.

2. If the above spring and block are cut in two equal halves speaker is set on one of the halves connected to one half spring and set in oscillation with same amplitude, find the maximum frequency of sound heard by observer rounded to closest integer value.

Space for Rough Work

Paragraph for Questions 3 and 4

The power radiated by sun is 4.0×10^{26} W. The earth orbits the sun in a nearly circular orbit of radius 1.5×10^{11} m. The earth's axis is tilted by 27° relative to the plane of the orbit, so sunlight does not strike perpendicularly. $\left[\cos 27^\circ = 0.9,\ \pi = \frac{22}{7}\right]$



3. Find the power (in W) that strikes a 0.75 m^2 patch of flat land at the equator at point Q ?

4. Find the average value of total energy density 'u' of the radiation incident at point Q on earth in μJm^{-3}.

Paragraph for Questions 5 and 6

A convex lens of focal length 20 cm is placed after a slit of width 0.6 mm. A plane wave of wavelength 6000 Å falls normally on the slit. A screen is placed at the focal plane of the convex lens.

5. Find the separation between the second minima on either side of the central maximum (in mm)

6. Find the ratio of the intensity of the principal maximum to the first maximum on either side of principal maximum. Consider secondary maximas to be placed right at the middle of minimas. [Take $\pi^2 = 10$]

Space for Rough Work

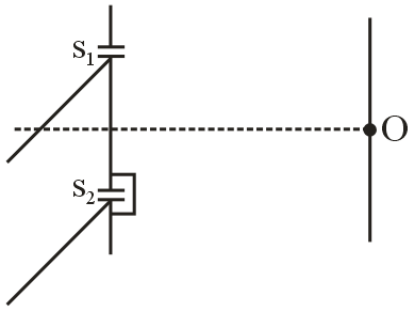
SECTION-II (ii) : (Maximum Marks: 12)

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7. In a young's double slit experiment a parallel beam of mono chromatic light of wavelength $\lambda = 600$ nm is incident on the plane of slits making an angle θ with the normal to the plane $(\sin\theta = 0.03)$. The separation between the slits is d = 0.1 mm. When a thin film is introduced in front of the the lower slit, three minimas and two maximas go past the center of the screen 'O' and the intensity at O reduces to one-fourth of the intensity before. If the refractive index of the material of thin film is 1.2, find the width of the film in μm.



8. Unpolarised light of intensity I_0 is incident on a polariser which is followed by a series of polarisers. Each polariser has its transmission axis rotated by 45° relative to the transmission axis of the previous polariser. What is the minimum number of polarisers needed for the light reaching the photo cell to have an intensity that is less than $\frac{I_0}{100}$.

9. A musical instrument has a spectrum of frequencies ranging from 100 Hz to 5000 Hz. The ear canal is a tube of length 3 cm and is open at one end and closed at the other. Find the frequency that will be best heard by ears in Hz. (take speed of sound = 330 m/s)

Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I (i) : (Maximum Marks: 24)

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1. Which point defect in crystals may reduce the crystal density ?

(A) Schottky defect

(B) Frenkel defect

(C) Metal excess defect

(D) Metal deficiency defect

Space for Rough Work

2. Under conditions of same temperature and pressure, mark **correct** relationship.

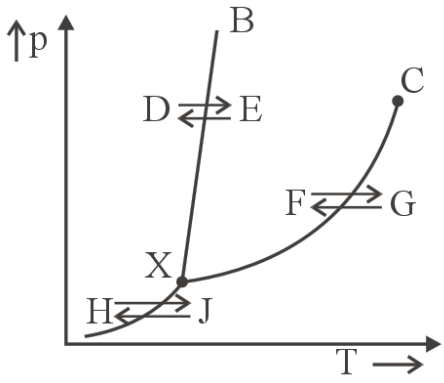
(A) Density of water vapour is more than N_2.

(B) Density of H_2 is more than water vapour.

(C) CO_2 is denser than air.

(D) Order of density : $H_2 < D_2 < T_2$

3. A generic phase diagram of a pure substance is drawn :



Choose **correct** statement :

(A) At C, surface tension of liquid becomes zero.

(B) At X, pure substance exists in solid, liquid and gaseous state simultaneously

(C) XB represents variation of boiling point of the substance, with pressure.

(D) At low T and high P, substance exhibits ideal gas behaviour

Space for Rough Work

4. Sodium hydroxide cannot be used as a primary standard for acid base titration because

(A) it is corrosive in nature and reacts with glass.

(B) the dissolution of sodium hydroxide in water is highly exothermic, hence its concentration changes on dissolution.

(C) it is hygroscopic and also reacts with atmospheric carbon dioxide.

(D) Hydroxides cannot be used as a primary standard.

5. The following equilibrium exists in saturated solution of NH_4Cl

$NH_4Cl_{(s)} \rightleftharpoons NH^+_{4(aq)} + Cl^-_{(aq)}$

$\Delta H_{25^\circ C} = 3.5$ kcal mol^{-1}

A change that will not shift the equilibrium to the right is

(A) decrease in temperature

(B) increase in temperature

(C) addition of NH_4Cl crystals to the reaction mixture.

(D) addition of NH_4OH solution to the reaction mixture.

6. Wavelength associated with electron in motion

(A) increases with decrease in the speed of electrons

(B) remains same irrespective of speed of electron

(C) decreases with increase in the speed of electron

(D) changes with the atomic number of atom to which it belongs

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

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- This section contains **TWO** paragraphs.
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Paragraph for Questions 7 and 8

The wave function for an atomic orbital of single electron atom or ion is

$\Psi(r, \theta, \phi) = \frac{2}{3}\left(\frac{Z}{3a_0}\right)^{\frac{3}{2}} (1-\sigma)(12-8\sigma+\sigma^2).\sigma.e^{-\frac{\sigma}{2}}\cos\theta$. Where $\sigma = \frac{2Zr}{na_0}$ $a_0 = 0.529$ Å.

All other parameters have their usual meaning.

7. $(n + \ell)$ Value of the given atomic orbital will be

(A) 2 (B) 3 (C) 4 (D) 6

8. Find the value of m_ℓ for the given orbital (Assuming z-axis is the internuclear axis)

(A) 0 (B) 2 (C) 1 (D) –2

Paragraph for Questions 9 and 10

The reaction : $A(g) \longrightarrow 2B(g) + C(\ell)$ obeys first order kinetics. 0.50 moles of A(g) is taken in an empty vessel of 16.4 L capacity and maintained at 27°C, at which A(g) start decomposing. In first 30 min, moles of B(g) becomes 0.75.

[Given : Vapour pressure of $C(\ell)$ at 27°C = 0.6 atm, R = 0.082 L-atm / K-mol]

9. Total pressure of system at t = 30 min. is :

(A) 1.875 atm (B) 1.3125 atm (C) 0.1875 atm (D) 1.0312 atm

10. The time at which first drop of liquid 'C' will form (log 2 = 0.3)

(A) 30 min (B) 35 min (C) 15 min (D) 60 min

Space for Rough Work

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Paragraph for Questions 1 and 2

Following equilibria are established on mixing two gases A_2 and C.

(i) $3A_{2(g)} \rightleftharpoons A_{6(g)}$, $K_p = 1.6\ atm^{-2}$

(ii) $A_{2(g)} + C_{(g)} \rightleftharpoons A_2C_{(g)}$

Given that the total pressure is 1.4 atm and partial pressure of A_6 is 0.2 atm,

If A_2 and C are mixed in 2 : 1 molar ratio, then

1. Calculate equilibrium partial pressure of C.

2. The value of K_p for the reaction (ii) (in atm^{-1}) is.

Paragraph for Questions 3 and 4

CO_2 gas (1.00 mole) at 400 K occupies 600 ml.

[For CO_2 ; a = 3.60 L^2atm mol^{-2}; b = 0.04 L mol^{-1}; R = 0.08 L atm k^{-1} mol^{-1}].

3. Calculate the pressure (in atm) exerted by the gas using ideal gas equation.

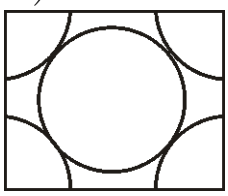
4. Calculate the pressure (in atm) exerted by the gas using Vander waal's gas equation.

Space for Rough Work

Paragraph for Questions 5 and 6

The given figure can be a plane projection in the cubic crystal packing of identical particles in a solids. The given figure may or may not be a square. Consider atoms to be perfectly circular/spherical as the case may be.

$\left(\text{Packing fraction} = \frac{\text{Packing efficiency}}{100}\right)$



5. Calculate packing efficiency, if the packing is in 2D.

6. Calculate maximum packing fraction, if the packing is in 3D.
(Express your answer as nearest integer)

Space for Rough Work

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7. In the reduction of nitric oxide, 50% of reaction was completed in 100 s, when initial pressure was 360 mm Hg and in 144 s, when initial pressure was 300 mm Hg. Find order of the reaction.

8. What is the pH at 25°C of the solution obtained by dissolving a 5.00 g tablet of aspirin in 0.500 L of water ? The 5 g tablet contains 0.30 g of aspirin ($HC_9H_7O_4$). The acid is monoprotic and $K_a = \left(\frac{10^{-3}}{3}\right)$ at 25°C. Write nearest integral value of pH. $\left(\sqrt{41} = 6.4,\ \sqrt{123} = 11.1\right)$
(log 3 = 0.48, log 2 = 0.30, log 7 = 0.85)

9. If pd versus p (where p in atm and d in g/L) is plotted for He gas at a particular temperature. If $\left[\frac{d}{dp}(pd)\right]_{p=8.21\ atm} = 5$, what is the temperature in Kelvin ? Assume ideal gas behaviour.
(R = 0.0821 L-atm/K-mol)

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I (i) : (Maximum Marks: 24)

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1. If value of definite integral $\int_0^{2\pi} \max\left(2\sin x + \cos 2x, \ \frac{8x - 15\pi}{\pi}\right) dx$ is 'A' then -

(where [.] → represent G.I.F.)

(A) $A > 0$ (B) $A < 0$

(C) $[A] = 0$ (D) $[A] = -2$

Space for Rough Work

2. Let us consider the function $f(x) = e^{-x} \cdot \sin x$, then

(A) Area bounded by the curve f(x) and x-axis between ordinate $x = -2\pi$ and $x = 0$ is $\left(\frac{e^{\pi}+1}{\sqrt{2}}\right)^2$

(B) Area bounded by the curve f(x) and x-axis between ordinate $x = -\pi$ and $x = \pi$ is $\frac{e^{\pi}-e^{-\pi}}{2}$

(C) $\int_{m\pi}^{(m+1)\pi} f(x)\,dx < \int_{n\pi}^{(n+1)\pi} f(x)\,dx$ where $(m, n \in I)$ and $(m < n)$

(D) $\int_{2m\pi}^{(2m+1)\pi} f(x)\,dx < \int_{2n\pi}^{(2n+1)\pi} f(x)\,dx$ where $(m, n \in I)$ and $(m > n)$

3. Consider differential equation $y^2\,(x \cdot e^y dy - e^y \cdot dx - x \cdot dx) + xy\left(xdy - \left(\frac{dy}{y} + \frac{dx}{x}\right)\right) = x^2y^2$

If $y(1) = 1$. If solution of differential equation is $\frac{1}{x}(h(y)) + \ln(g(x,\ y)) = x + e$, then

(A) $h(2) = \frac{2 \cdot e^2 + 1}{2}$

(B) $g(x, y) = 5$ represent a parabola passing through origin

(C) $h(x)$ is an increasing function for $x \in [1, \infty)$

(D) $g(x, y) = 5$ represent a straight line

Space for Rough Work

4. $\lim\limits_{n\to\infty} \dfrac{(1^{\lambda}+2^{\lambda}+3^{\lambda}+\ldots\ldots+n^{\lambda})}{(1^4+2^4+\ldots\ldots+n^4)(1^3+2^3+\ldots\ldots+n^3)} = F(\lambda),\ \lambda \in N$ then

(A) $F(\lambda)$ is finite for $\lambda \le 8$

(B) $F(8) = \dfrac{20}{11}$

(C) $F(8) = \dfrac{20}{9}$

(D) $F(7) = 0$

5. If the least value of area bounded by $y = x^2 - 3$ and $y = kx + 2$ be A, then the value of $\dfrac{3A}{10\sqrt{5}}$ is

(A) less than 2

(B) greater than 5

(C) less than 4

(D) greater than 1

6. Let f(x) be a polynomial of degree 3 such that f(0) = 1, f(1) = 2, x = 0 is a critical point but f(x) does not have local extremum at x = 0, then $\int \dfrac{f(x)}{\sqrt{x^2+7}}dx$ is equal to

(A) $\dfrac{\sqrt{x^2+7}}{3}(x^2-14) + \ln\left(x+\sqrt{x^2+7}\right) + C$

(B) $\dfrac{\sqrt{x^2+7}}{3}(x^2-14) - \ln\left(\sqrt{x^2+7}-x\right) + C$

(C) $\ln\left(x+\sqrt{x^2+7}\right) + \sqrt{x^2+7}\,(x^2-7) + C$

(D) $\ln\left(x-\sqrt{x^2+7}\right) + \sqrt{x^2+7}\,(x^2-7) + C$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct option is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : –1 In all other cases.

Paragraph for Questions 7 and 8

Let f be a monotonic and differentiable function satisfying $\int_0^{f(x)} f^{-1}(t)dt - \int_0^{x} (\cos t - f(t))\,dt = 0$ and $f(0) = 1$.

7. The number of solution(s) of the equation $\begin{vmatrix} f(2x) & f(x) \\ \sin x & 2 \end{vmatrix} = 0$ in $(0, 2\pi)$ is

(A) 3 (B) 6 (C) 4 (D) 5

8. If the value of $\int_0^{\pi/2} f(x)dx = \lambda$, then $[|\lambda|]$ is equal to, [ ] = G.I.F.

(A) 2 (B) 3 (C) 4 (D) 1

Space for Rough Work

Paragraph for Questions 9 and 10

Let : $\int_0^{\pi} \frac{x}{1-\cos\phi.\sin x}dx = f(\pi - \phi)$

9. Then the value of $f\left(\frac{\pi}{6}\right)$ is

(A) π^2

(B) $\frac{\pi^2}{2}$

(C) $\frac{\pi^2}{3}$

(D) $\frac{\pi^2}{4}$

10. Then the value of $\int_0^{\frac{\pi}{2}} f^2(x)\,dx$ is

(A) $\pi^3 \ln 2$

(B) $2\pi^3 \ln 2$

(C) $\frac{\pi^3}{2}\ln 2$

(D) $3\pi^3\ln 2$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;
Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Let f(x) be a non constant function such that $f(x) = x^2 - \int_0^1 (f(t) + x)^2 dt$

$a = \int_0^{\infty} \frac{\tan^{-1}(x)}{x(1+x^2)} dx$ $\quad$ $b = \int_0^1 \frac{5x^8 + 3x^4 + 1}{\sqrt{x^8 + x^4 + 1}} dx$

$c = \int_0^2 \sqrt{x^3 + 1}\ dx + \int_1^3 (x^2 - 1)^{\frac{1}{3}} dx$ $\quad$ $d = \int_0^1 \frac{3x^2 + 9x + 7}{(x^2 + 3x + 2)^3} dx$

1. If A is the area of region bounded by f(x) and x axis from $x = \frac{2a}{\pi \ln 2}$ to $x = b^2$, then $\frac{A}{4}$ is

2. If S area of region bounded by $f^{-1}(x)$ and x axis from $x = \frac{72d}{11}$ to x = c, then $\frac{[S]}{2}$ is (where [.] GIF)

Space for Rough Work

Paragraph for Questions 3 and 4

$I = \int f\left(x^m (ax^n + b)^{r/s}\right) dx$; r, s ∈ I & m, n ∈ Q.

If $\frac{m+1}{n}$ = integer, then we can solve by substituting $ax^n + b = t^2$ and if $\frac{m+1}{n} \neq$ integer but $\frac{m+1}{n} + \frac{r}{s}$ = integer; we can substitute $a + bx^{-n} = t^s$; it converts function into integrable form.

3. If $\int \frac{dx}{\sqrt{a^{5x}}\sqrt[4]{(a^{2x} + a^{-2x})^3}} = \frac{-1}{\ln a} g(x) + C$, then $g\left(-\frac{1}{4}\log_a 15\right)$ is equals -

4. $\int \sqrt{\text{cosec}\, x}\left(2\sqrt[3]{\text{cosec}\, x} + 1\right)^{\frac{3}{2}} \cdot \cos x\, dx = pt^3 + qt + r\sqrt{2}\ln\left|\frac{t-\sqrt{2}}{t+\sqrt{2}}\right| + C$ (where $t = \sqrt{2 + \sin^{\frac{1}{3}} x}$)

then p +q – r is equal

Paragraph for Questions 5 and 6

Let f(x) be a polynomial of degree 3 passing through origin and have a local maximum at $x = \frac{-1}{\sqrt{10}}$. Also $f'(x)$ has a local minimum at x = 0 and f(1) = 7.

5. The value of f(3) is equal to

6. Consider a real valued function y = g(x) satisfying the differential equation $\frac{dy}{dx} + f'(x) \cdot y = (f(x) + 1) f'(x)$ such that g(0) = 1 and $g(1) = \alpha + e^{-\beta}$. Then (α + β) is equal to

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.
Zero Marks : 0 In all other cases

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7. The ratio of the areas in which the curve $y = \left[\frac{x^3}{101} + \frac{x}{36}\right]$ divides the circle $x^2 + y^2 - 4x + 2y + 1 = 0$, (where [.] denotes greatest integer function) is $\frac{k\pi - 3\sqrt{3}}{m\pi + 3\sqrt{3}}$ then value of $|k + m|$ _____

8. Consider the family of curve $y = f(x)$ whose differential equation is given by $x\frac{dy}{dx} + y = \frac{2\cos x + 3}{(2 + 3\cos x)^2}$.
If $f\left(\frac{\pi}{2}\right) = \frac{2}{\pi}$, then value of $\left[\pi f\left(\frac{\pi}{3}\right)\right]$ is (where [.] is GIF)

9. Let f be a continuous function on $[0, \infty)$ such that $\lim\limits_{x\to\infty}\left(f(x) + \int\limits_0^x f(t)\,dt\right)$ exists ; then find $\lim\limits_{x\to\infty}\left(f(x) + x\sin\frac{1}{x}\right)$

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE .....

FORM NO. .....

I have read all the instructions and shall abide by them. _____ Signature of the Candidate	I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same. _____ Signature of the Invigilator

Space for Rough Work

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2022 - 2023)

JEE (Advanced)
PART TEST
18-12-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS, TNAS, I(A) & LIVE-I(A)] (SCORE-I)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,C,D	B,D	A,B,D	B,C,D	B,C,D	C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	C	A	D	A		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.18 to 2.21	602.00	954.54 to 954.55	4.71	0.80	22.50
SECTION-II (ii)	Q.	7	8	9			
	A.	8	7	2750			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,C,D	C,D	A,B	C	A,C,D	A,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	D	A	A	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.40	1.50	53.33	47.14	78.50 to 78.60	0.74
SECTION-II (ii)	Q.	7	8	9			
	A.	3	3	160			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,C	A,D	A,C,D	A,C,D	C,D	A,B
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	D	C	A		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	1.50	4.50	2.00	8.00	261.00	14.00
SECTION-II (ii)	Q.	7	8	9			
	A.	12	2	1			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,C,D)

$$I = \frac{P}{A} = \frac{1.77 \times 10^3\,W}{0.2 m^2} = 8.85 \times 10^3\,W/m^2$$

Peak intensity is double average intensity

$$I = \frac{1}{2}\varepsilon_0 E_0^2 C$$

$$E_0^2 = \frac{2I}{\varepsilon_0 C} = \frac{2 \times 8.85 \times 10^3}{8.85 \times 10^{-12} \times 3 \times 10^8}$$

$$= 6.66 \times 10^6\,N^2/C^2$$

$$\left(\frac{1}{2\mu_0}B_0^2\right) = \frac{I}{C} = \frac{8.85 \times 10^3}{3 \times 10^8}$$

$$= 2.95 \times 10^{-5}\,J/m^3$$

2. Ans (B,D)

Theory based questions

3. **Ans (A,B,D)**

y (I st minima) = 225 W

$W \sin\theta = \lambda$

$\rightarrow \sin\theta = \frac{\lambda}{W} = \tan\theta = \frac{225}{18000} = \frac{1}{80}$

For y(2nd minima),

$W \sin\theta = 2\lambda \rightarrow \tan\theta = \frac{1}{40}$

$I = I_0\left[\frac{\sin\left(\frac{\phi}{2}\right)}{\left(\frac{\phi}{2}\right)}\right]^2,$

$\Delta x = W \sin\theta = \frac{W}{480} = \frac{\lambda}{6} \rightarrow \phi = \frac{2\pi}{\lambda}(\Delta x)$

$\phi = \frac{\pi}{3}$

4. **Ans (B,C,D)**

Theory based questions

5. **Ans (B,C,D)**

$L_1 = 10\log_{10}\frac{I_1}{I_0}$

$L_2 = 10\log_{10}\frac{I_2}{I_0}$

$\frac{I_1}{I_2} = 10^{0.4} = \left(10^{0.2}\right)^2 = 2.56$

$\frac{I_1}{I_2} = \left(\frac{r_2}{r_1}\right)^2, \quad \frac{r_2}{r_1} = 10^{0.2} = 1.6$

$r_2 = 1.6\, r_1$ also $r_2 - r_1 = 3$

$1.6r_1 - r_1 = 3 \Rightarrow 0.6r_1 = 3$

$\Rightarrow r_1 = 5$ m

$r_2 = 1.6 \times 5 = 8$ m.

6. **Ans (C)**

Acceleration of the pulse remains constant in all the three strings.

$\frac{\lambda_B}{\lambda_C} = \frac{v_B}{v_C} = \sqrt{\frac{T_B}{T_C}} \times \sqrt{\frac{\mu_C}{\mu_B}} = \sqrt{\frac{5}{3}} \times \sqrt{\frac{3}{1}} = \sqrt{5} : 1$

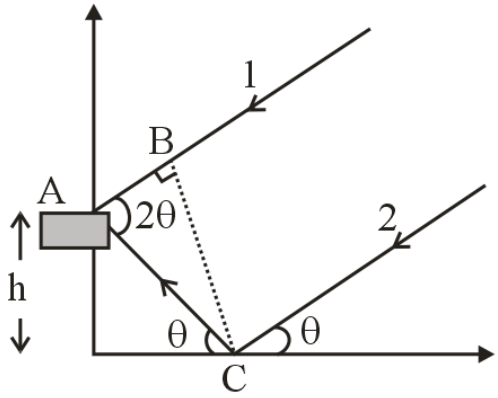
PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (C)**

The signal detected by the detector A, is a result of interference of two rays the ray 1 incident directly from the star and ray 2, reflected from sea surface.

The phase of second ray is shifted by π due to the reflection by a denser medium.



Therefore the path difference between the two rays is :

$\Delta x = AC + \frac{\lambda}{2} - AB = \frac{h}{\sin\theta} + \frac{\lambda}{2} - \frac{h}{\sin\theta}\cos 2\theta$

$= \frac{\lambda}{2} + \frac{h}{\sin\theta}(1 - \cos 2\theta) = \frac{\lambda}{2} + 2h\sin\theta$

Condition for 2nd maxima is :

$\frac{\lambda}{2} + 2h\sin\theta = 2\lambda$

$\sin\theta = \frac{3\lambda}{4h} = \frac{3}{4} \times \frac{21\text{cm}}{210\text{cm}} = 0.075$

So N = 75

8. **Ans (A)**

$$E_{max} = E_i + E_r = E_i + E_i\left(\frac{n - \cos i}{n + \cos i}\right)$$

Since $i = \frac{\pi}{2} - \theta$

$$E_{max} = E_i\left(1 - \frac{n - \sin\theta_{max}}{n + \sin\theta_{max}}\right)$$

$$= E_i \frac{2n}{n + \sin\theta_{max}}$$

$$\frac{I_{1st\ maxima}}{I_{next\ minima}} = \left(\frac{n^2}{\sin^2\alpha_{min}}\right)\left(\frac{n + \sin\alpha_{min}}{n + \sin\alpha_{max}}\right)^2$$

For 1st maxima

$$\rightarrow \frac{\lambda}{2} + 2h\sin\theta = \lambda$$

$$\rightarrow \sin\theta = \frac{\lambda}{4h} = \frac{21}{4 \times 210}$$

$$\rightarrow \sin\theta_{max} = 0.025$$

For next minima

$$\rightarrow \frac{\lambda}{2} + 2h\sin\theta = \frac{3}{2}\lambda$$

$$\rightarrow \sin\theta = \frac{\lambda}{2h}$$

$$\rightarrow \sin\theta_{min} = 0.05$$

$$\frac{I_{max}}{I_{min}} = \left(\frac{9}{0.05}\right)^2 \times \left(\frac{9 + 0.05}{9 + 0.025}\right)^2$$

Ans. 3.25 - 3.26

9. **Ans (D)**

The frequencies are in the ratio of 5 : 7 : 9.

Fixed at one end & free at other.

The fundamental frequency is the H.C.F. of 175, 245 and 315, so $f_0 = 35$ Hz

175 Hz is 5th Harmonic $\rightarrow \lambda_1 = \frac{2L}{5}$

245 Hz is 7th Harmonic $\rightarrow \lambda_2 = \frac{2L}{7}$

315 Hz is 9th Harmonic $\rightarrow \lambda_3 = \frac{2L}{9}$

$\alpha + \beta + \gamma = 11.68$ m.

10. **Ans (A)**

With two nodes and antinodes, it is 3rd harmonic

$$\lambda = \frac{4L}{3} = 17.16\text{m}$$

$v = \lambda f = 600.6$ m/s

$T = \mu v^2 = 360.72$ N

PART-1 : PHYSICS

SECTION-II (i)

1. **Ans (2.18 to 2.21)**

$$f_h = \left(\frac{v}{v - v_s}\right) f,\ f_L = \left(\frac{v}{v + v_s}\right) f$$

$$f_h - f_L = \left[\frac{v}{v - v_s} - \frac{v}{v + v_s}\right] f$$

$$= \left[\frac{1}{1 - \frac{v_s}{v}} - \frac{1}{1 + \frac{v_s}{v}}\right] f$$

$$= \left[\left(1 - \frac{v_s}{v}\right)^{-1} - \left(1 + \frac{v_s}{v}\right)^{-1}\right] f$$

Since difference is small $\frac{v_s}{v} << 1$

So,

$$f_h - f_L = \left[\left(1 + \frac{v_s}{v}\right) - \left(1 - \frac{v_s}{v}\right)\right] f = \frac{2v_s}{v} f$$

$$\rightarrow 2 = \frac{2v_s}{330} \times 600$$

$$\rightarrow v_s = \frac{330}{600} = \frac{11}{20}\ \text{m/s.}$$

$$v_s = A\omega = 2\pi f A = 8 \times \frac{22}{7} \times A$$

$$A = \frac{7}{320}\text{m} = 2.19$$

2. **Ans (602.00)**

$\omega = \sqrt{\frac{k}{m}}$, Half length spring has double k and mass is also halved, So $\omega' = 2\omega$,

$v_s' = 2v_s$ previous $\frac{v_s}{v} = \frac{1}{600}$

New

$$\frac{v_s'}{v} = \frac{2}{600},$$

$$f' = \left(1 + \frac{v_s'}{v}\right) 600 = 602\text{Hz}$$

3. **Ans (954.54 to 954.55)**

$$I = \frac{P_0}{4\pi r^2}$$

$$P = IA\cos\theta$$

$$P = \frac{P_0}{4\pi r^2} A\cos\theta$$

$$= \frac{4 \times 10^{26} \times 0.75 \times 0.9}{4 \times \frac{22}{7} \times 2.25 \times 10^{22}}$$

$$= \frac{2.1}{22} \times 10^4$$

$$= 954.54$$

4. **Ans (4.71)**

Intensity = Total energy density × speed of radiation

$$U = \frac{I}{C} = \frac{P_0}{4\pi r^2 C}$$

$$= \frac{4.0 \times 10^{26}}{4 \times \frac{22}{7} \times (1.5 \times 10^{11})^2 \times 3 \times 10^8}$$

$$= \frac{7 \times 10^4}{22 \times 2.25 \times 3 \times 10^8}$$

$$= \frac{700}{148.5} \times 10^{-6}$$

$$= 4.71 \times 10^{-6}\ Jm^{-3}$$

5. **Ans (0.80)**

For second minima $a \sin\theta = 2\lambda$

$$\sin\theta = \frac{2\lambda}{a} = \frac{2 \times 6 \times 10^{-7}}{0.6 \times 10^{-3}}$$

$$\sin\theta = 2 \times 10^{-4} = \tan\theta$$

The diffraction pattern is forming on the screen at a distance of 20 cm distance y from central maxima on the screen.

so, $\tan\theta = \frac{y}{20cm} = 2 \times 10^{-4}$

$y = 40 \times 10^{-4}$ cm.

Separation between two minimas

$= 2y = 80 \times 10^{-4}$ cm.

= 0.08 mm.

6. **Ans (22.50)**

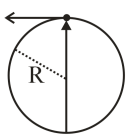
At central maxima all the waves originating from the plane wave front superpose constructively. Let the resultant amplitude be A

$\xrightarrow[A]{}$

At 1st maxima path difference between one end of wave front and the other is $\frac{3\lambda}{2}$, i.e. phase difference of 3π

So resultant amplitude is 2R

$$3\pi R = A \Rightarrow R = \frac{A}{3\pi} \qquad A' = \frac{2A}{3\pi}$$



Ratio of intensity

$$= \frac{A^2}{\left(\frac{2A}{3\pi}\right)^2} = \frac{9\pi^2}{4} = \frac{90}{4} = 22.5$$

PART-1 : PHYSICS

SECTION-II (ii)

7. **Ans (8)**

Path difference at source

$= d \sin\theta = 0.1 \times 0.03mm = 3 \times 10^{-6} m$

$= 5 \times 0.6 \times 10^{-6}$ m

$= 5\lambda$

Since 3 minimas and two maximas have gone past O

So new path difference due to thin film $\frac{8\lambda}{3}$

$$\frac{8}{3}\lambda = (\mu - 1)\, t$$

$$t = \frac{8}{3}\frac{\lambda}{(\mu - 1)} = \frac{8 \times 0.6\mu m}{3 \times (1.2 - 1)} = 8\mu m$$

8. **Ans (7)**

Intensity after 1st polarioid is $\frac{I_0}{2}$

for each subsequent polariser $I = \frac{I_0}{2}(\cos^2\theta)^n$

Where n is the number of subsequent polarisers.

$$I = \frac{I_0}{2}\left(\frac{1}{2}\right)^n = \frac{I_0}{2^{n+1}} < \frac{I_0}{100}$$

Minimum value of n satisfying this is n = 6

So total polariser is n + 1 = 7

9. **Ans (2750)**

When frequency of sound matches with frequency of natural vibration of air column, resonance happens Which causes the amplitude to become large. So resonant frequencies will be heard best

$$f_n = (2n - 1)\frac{v}{4\ell}$$

$$f_1 = \frac{v}{4\ell} = \frac{330\ ms^{-1}}{4 \times 0.03} = 2750$$

$$f_2 = 3\frac{v}{4L} = 8250$$

So in the given range the best heard frequency will be 2750 Hz.

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (A,C,D)**

$\rho_{exp.} < \rho_{theo.}$

2. **Ans (C,D)**

$d = \frac{PM}{RT}$

3. **Ans (A,B)**

Factual

4. **Ans (C)**

NaOH reacts with CO_2 & H_2O present in air.

5. **Ans (A,C,D)**

Apply L.C.P.

6. **Ans (A,C)**

$\lambda = \frac{h}{mv}$;

PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (D)**

$\Psi(r,\theta,\phi) \alpha r^{\ell}$

$\therefore \quad \ell = 1$

No. of radial node = 3

Radial node at :

$(1-\sigma)(12-8\sigma+\sigma^2) = 0$

$\therefore \quad n = 5$

$\therefore \quad (n+\ell) = 6$

8. **Ans (A)**

Angular Node at :

$\cos\theta = 0 \Rightarrow \theta = \frac{\pi}{2} \Rightarrow$ xy plane

$\therefore$ Orbital is p_z

$\therefore \quad m_{\ell} = 0$

10. **Ans (B)**

For first drop of C(ℓ) formed, pressure due to vapours of C should be 0.6 atm. For it, moles of C formed should be $\frac{0.6 \times 16.4}{0.082 \times 300} = 0.4$.

PART-2 : CHEMISTRY

SECTION-II (i)

1. **Ans (0.40)**

$3A_{2(g)} \rightleftharpoons A_{6(g)}$

$K_p = 1.6 = \frac{0.2}{p_{A_2}^3}$

$\Rightarrow p_{A_2} = 0.5$ atm

Let the partial pressure of A_2 and C before the start of reaction be a and a/2 atm respectively.

$A_{2(g)} + C_{(g)} \rightleftharpoons A_2C_{(g)}$

If x moles of A_2 reacts to form x moles of A_2C.

$1.4 = 0.5 + 0.2 + p_{A_2C} + p_C$

$\Rightarrow \quad p_{A_2C} + p_C = 0.7$

$a/2 - x + x = 0.7$

$a = 1.4$ atm

Now amount of A_2 Consumed in first reaction

$= 3 \times 0.2 = 0.6$ atm

2. **Ans (1.50)**

Amount of A_2 consumed in second reaction

$= 1.4 - 3 \times 0.2 - 0.5 = 0.3$ atm

$\Rightarrow \quad p_{A_2C} = 0.3$atm

$p_c = 0.7 - 0.3 = 0.4$ atm

$K_p = \frac{0.3}{0.4 \times 0.5} = 1.5\,atm^{-1}$

3. **Ans (53.33)**

$P = \frac{nRT}{V}$

$= \frac{1.00 \times 0.08 \times 400}{0.60} = 53.33$ atm.

4. **Ans (47.14)**

$\left(P + \frac{n^2a}{V^2}\right)(V - nb) = nRT$

P= 47.14

5. **Ans (78.50 to 78.60)**

$\eta = \frac{2 \times \pi R^2}{a^2}$

$a\sqrt{2} = 4R$

$\eta = \frac{2 \times \pi R^2}{4 \times 4 \times R^2 \times \frac{1}{2}} = \frac{\pi}{4}$

6. **Ans (0.74)**

FCC ; $\eta = 74\%$

Ans : 74

PART-2 : CHEMISTRY

SECTION-II (ii)

7. **Ans (3)**

$t_{\frac{1}{2}} \propto a^{1-n}$

$n = 3$

8. **Ans (3)**

$HAcs_{(aq)} \rightleftharpoons H^{+}_{(aq)} + Acs^{-}_{(aq)}$

c

$c(1.\alpha)$ $\quad c\alpha \quad c\alpha$

$k_a = \dfrac{c\alpha^2}{1-\alpha}$

$c = \dfrac{\frac{0.325}{179}}{0.5} = 3.6 \times 10^{-3}$

$\Rightarrow 10\alpha^2 + \alpha - 1 = 0$

$\alpha = 0.27$

$[H^+] = 0.27 \times 3.6 \times 10^{-3}$

$pH = 3.0$

9. **Ans (160)**

As we know

$pV = nRT$

$p = \dfrac{n}{V}RT$

$\Rightarrow \quad p = \dfrac{\left(\frac{w}{M}\right)RT}{V} \qquad \therefore n = \dfrac{w}{M}$

$\Rightarrow \quad p = \dfrac{w.RT}{V.M} = \dfrac{w}{V} \times \dfrac{RT}{M}$

$\Rightarrow \quad p = d \cdot \dfrac{RT}{M} \qquad \left(\because \dfrac{w}{V} = d\right)$

$\Rightarrow \quad d = \dfrac{pM}{RT}$

Multiply both sides by p

$\Rightarrow \quad pd = p \times \dfrac{pM}{RT} = \dfrac{p^2M}{RT} \qquad ....(i)$

On differentiating Eq. (i) w.r.t. p, we will get

$\dfrac{d(p.d)}{dp} = \dfrac{d}{dp}\left(\dfrac{p^2M}{RT}\right) = \dfrac{2pM}{RT}$

$5 = \dfrac{2pM}{RT}$

$T = \dfrac{2pM}{5R} = \dfrac{2 \times 8.21 \times 4}{5 \times 0.0821}$ [for He, M = 4]

$T = 160$ K

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (A,C)**

$y = 2\sin x + \cos 2x$

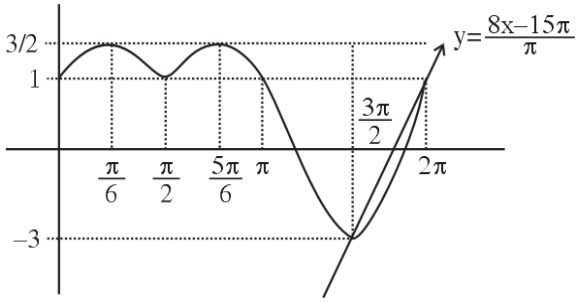
$y' = 2\cos x - 2\sin 2x$

$\dfrac{dy}{dx} = 2\cos x(1 - 2\sin x)$

$\dfrac{dy}{dx} = 0 \Rightarrow x = \dfrac{\pi}{6}, \dfrac{\pi}{2}, \dfrac{5\pi}{6}, \dfrac{3\pi}{2}$

$\dfrac{dy}{dx} \Rightarrow$ + | − | + | − | +

$\dfrac{\pi}{6} \quad \dfrac{\pi}{2} \quad \dfrac{5\pi}{6} \quad \dfrac{3\pi}{2}$



$A = \displaystyle\int_0^{\frac{3\pi}{2}} (2\sin x + \cos 2x)\,dx + \int_{\frac{3\pi}{2}}^{2\pi} \left(\dfrac{8x - 15\pi}{\pi}\right) dx$

$A = \left|-2\cos x + \dfrac{\sin 2x}{2}\right|_0^{\frac{3\pi}{2}} + \dfrac{1}{\pi}\left(4x^2 - 15\pi x\right)_{\frac{3\pi}{2}}^{2\pi}$

$A = (2) + \dfrac{1}{\pi}\left\{\left(16\pi^2 - 30\pi^2\right) - \left(9\pi^2 - \dfrac{45\pi^2}{2}\right)\right\}$

$A = 2 + \dfrac{1}{\pi}\left\{-14\pi^2 - 9\pi^2 + \dfrac{45\pi^2}{2}\right\}$

$A = 2 + \dfrac{1}{\pi}\left(-23\pi^2 + \dfrac{45\pi^2}{2}\right)$

$A = 2 + \dfrac{1}{\pi}\left(-\dfrac{\pi^2}{2}\right)$

$A = 2 - \dfrac{\pi}{2}$

2. **Ans (A,D)**

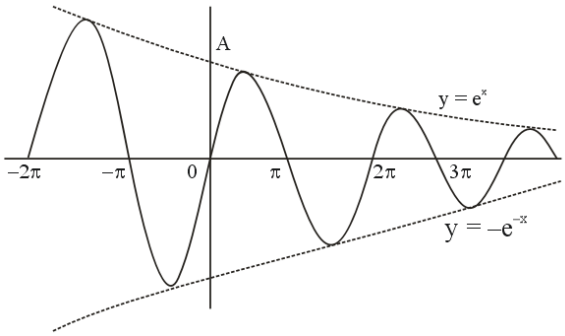
As we know $-1 \leqslant \sin x \leqslant 1$

$y = e^{-x} \sin x$

$\Rightarrow \quad -e^{-x} \leqslant y \leqslant e^{x}$

then y is bounded between

$y = -e^{-x}$ and $y = e^{-x}$



(A) Area bounded by curve and x-axis

between x = –2π and x= 0

$$A = \int_{-2\pi}^{-\pi} e^{-x} \sin x \, dx + \left| \int_{-\pi}^{0} e^{-x} \sin x \, dx \right|$$

$$A = \left(\frac{-e^{-x}(\sin x + \cos x)}{2} \right)_{-2\pi}^{-\pi} + \left| \left(\frac{-e^{-x}(\sin x + \cos x)}{2} \right)_{-\pi}^{0} \right|$$

$$= \left(\frac{e^{\pi} + 1}{\sqrt{2}} \right)^2$$

(B) Area bounded by curve and x axis between

ordiante x = –π and x = π

$$A = \left| \int_{-\pi}^{0} f(x) \, dx \right| + \int_{0}^{\pi} f(x) \, dx$$

$$A = \frac{e^{\pi} + e^{-\pi} + 2}{2}$$

from graph we can say that option (C) is not

always true but option (D) is always true

3. **Ans (A,C,D)**

$$y^2 (x \cdot e^y dy - e^y \cdot dx - x \cdot dx)$$

$$+xy \left(xdy - \left(\frac{dy}{y} + \frac{dx}{x} \right) \right) = x^2 y^2$$

$$\frac{x \cdot e^y \, dy - e^y \, dx}{x^2} + \frac{x \, dy - y \, dx}{xy} - \frac{(xdy + ydx)}{x^2 y^2} = 1$$

$$d\left(\frac{e^y}{x} \right) + d\left(\ln\left(\frac{y}{x} \right) \right) + d\left(\frac{1}{xy} \right) = 1$$

$$\frac{e^y}{x} + \ln\left(\frac{y}{x} \right) + \frac{1}{xy} = x + C$$

$\because \quad y(1) = 1$

$$\frac{e}{1} + o + 1 = 1 + C$$

$$C = e$$

$$\frac{e^y}{x} + \ln\left(\frac{y}{x} \right) + \frac{1}{xy} = e + x$$

$$\frac{1}{x}\left(e^y + \frac{1}{y} \right) + \ln\left(\frac{y}{x} \right) = e + x$$

$$h(y) = e^y + \frac{1}{y}$$

$$g(x, \; y) = \frac{y}{x}$$

$$h(2) = \frac{2 \cdot e^2 + 1}{2}$$

$$h'(x) = e^x - \frac{1}{x^2} > 0, \quad \forall x \in [1, \; \infty)$$

$$\Rightarrow \quad \frac{y}{x} = 5$$

$\Rightarrow$ y = 5x represent a st. line passing through origin

4. **Ans (A,C,D)**

$$F(\lambda) = \lim_{n \to \infty} \frac{\sum_{r=1}^{n} r^{\lambda}}{\left(\sum_{r=1}^{n} r^4 \right) \left(\sum_{r=1}^{n} r^3 \right)}$$

$$= \lim_{n \to \infty} \frac{\left(\frac{1}{n} \sum_{r=1}^{n} \left(\frac{r}{n} \right)^{\lambda} \right) \cdot n^{\lambda + 1}}{\left(\frac{1}{n} \sum_{r=1}^{n} \left(\frac{r}{n} \right)^4 \right) \left(\frac{1}{n} \sum_{r=1}^{n} \left(\frac{r}{n} \right)^3 \right) \cdot n^9}$$

$$\lim_{n \to \infty} \frac{\int_0^1 x^{\lambda} dx}{\left(\int_0^1 x^4 dx \right) \left(\int_0^1 x^3 dx \right)} \cdot n^{\lambda - 8}$$

$$\Rightarrow \lim_{n \to \infty} \frac{20}{(\lambda + 1)} \cdot \frac{1}{n^{8 - \lambda}} = F(\lambda)$$

5. **Ans (C,D)**

$y = x^2 - 3$ ....(1)

$y = kx + 2$ ....(2)

solving (1) & (2)

$x^2 - kx - 5 = 0$

Let α and β are roots of this then

$\alpha + \beta = k$ and $\alpha\beta = -5$

Required area

$$A = \left[\int_{\alpha}^{\beta} (x^2 - 3) - (kx + 2)\right] dx$$

$$= \left[\frac{x^3}{3} - \frac{kx^2}{2} - 5x\right]_{\alpha}^{\beta}$$

$$= \left|\frac{1}{3}[\beta^3 - \alpha^3] - \frac{k}{2}[\beta^2 - \alpha^2] - 5(\beta - \alpha)\right|$$

$$A = \left|(\beta - \alpha)\left(\frac{\alpha^2 + \beta^2 + \alpha\beta}{3} - \frac{k}{2}(\beta + \alpha) - 5\right)\right|$$

Putting the values of $\alpha + \beta$ and $\alpha\beta$

$$A = \frac{1}{6}(k^2 + 20)^{\frac{3}{2}}$$

for least value of A, k = 0

Least value of $A = \frac{1}{6}(20)^{\frac{3}{2}} = \frac{20\sqrt{5}}{3}$

6. **Ans (A,B)**

Since x = 0 is critical point of f(x) but f(x) does not have local extrema at x = 0

so $f'(0) = 0$ and $f''(0) = 0$

$\Rightarrow f'(x) = a(x - 0)^2$

$$f(x) = \frac{ax^3}{3} + C$$

$\Rightarrow f(0) = 1,\ f(1) = 2$

$$f(x) = x^3 + 1$$

$$\Rightarrow \int \frac{x^3 + 1}{\sqrt{x^2 + 7}} dx = \int \frac{x^3 dx}{\sqrt{x^2 + 7}} + \int \frac{dx}{\sqrt{x^2 + 7}}$$

$$\Rightarrow \frac{(x^2 + 7)^{\frac{3}{2}}}{3} - 7(x^2 + 7)^{\frac{1}{2}} + \log\left(x + \sqrt{x^2 + 7}\right)$$

$$\Rightarrow \frac{\sqrt{x^2 + 7}}{3}[x^2 - 14] + \log\left(x + \sqrt{x^2 + 7}\right) + C$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (A)**

$$\int_0^{f(x)} f^{-1}(t)dt = \int_0^{x} (\cos(t) - f(t))dt$$

$$f^{-1}(f(x)) \cdot f'(x) = \cos x - f(x)$$

$$xf'(x) = \cos x - f(x)$$

$$xf'(x) + f(x) = \cos x$$

$$\frac{d}{dx}[xf(x))] = \cos x$$

$$xf(x) = \sin x + c$$

but f(0) = 1, c = 0

$$\Rightarrow f(x) = \frac{\sin x}{x}$$

$$\begin{vmatrix} f(2x) & f(x) \\ \sin x & 2 \end{vmatrix} = 0$$

$$2f(2x) - f(x) \cdot \sin x = 0$$

$$2 \cdot \frac{\sin 2x}{2x} - \frac{\sin x}{x} \cdot \sin x = 0$$

$$\sin x\,[2\cos x - \sin x] = 0$$

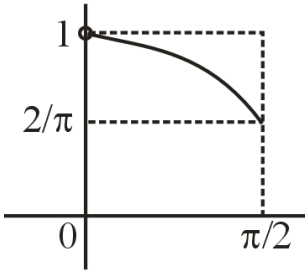
For $x \in (0, 2\pi)$

$\sin x = 0, \Rightarrow x = \pi$

$\tan x = 2$ two solution

total number of solutions = 3

8. **Ans (D)**



$$1 < \int_0^{\pi/2} \frac{\sin x}{x} < \frac{\pi}{2}$$

9. **Ans (C)**

Let $\int_0^{\pi} \frac{x}{1-\cos\phi.\sin x}dx = f(\pi-\phi)$ ....(1)

$$I = \int_0^{\pi} \frac{\pi - x}{1-\cos\phi\cdot\sin(\pi-x)}dx \quad ...(2)$$

Adding (1) & (2)

$$2I = \pi\cdot\int_0^{\pi}\frac{dx}{1-\cos\phi\cdot\sin x}$$

$$I = \frac{\pi}{2}\cdot\int_0^{\pi}\frac{dx}{1-\cos\phi\cdot\frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}}$$

$$= \frac{\pi}{2}\int_0^{\pi}\frac{\sec^2\frac{x}{2}\cdot dx}{1+\tan^2\frac{x}{2}-2\cos\phi\cdot\tan\frac{x}{2}}$$

Putting $\tan\frac{x}{2} = t$

$$\frac{1}{2}\sec^2\frac{x}{2}\cdot dx = dt$$

$$I = \pi\int_0^{\infty}\frac{dt}{1+t^2-2t\cos\phi}$$

$$= \pi\int_0^{\infty}\frac{dt}{(t-\cos\phi)^2+\sin^2\phi}$$

$$= \pi\left[\frac{1}{\sin\phi}\cdot\tan^{-1}\left(\frac{t-\cos\phi}{\sin\phi}\right)\right]_0^{\infty}$$

$$I = \pi\frac{1}{\sin\phi}\left[\frac{\pi}{2}-\left(\tan^{-1}(-\cot\phi)\right)\right]$$

$$= \frac{\pi}{\sin\phi}\cdot\left[\frac{\pi}{2}+\frac{\pi}{2}-\phi\right]$$

$$= \frac{\pi(\pi-\phi)}{\sin\phi} = f(\pi-\phi) \text{ (As given)}$$

$$\Rightarrow f(x) = \pi\frac{x}{\sin x}$$

$$f\left(\frac{\pi}{6}\right) = \frac{\pi\cdot\frac{\pi}{6}}{\frac{1}{2}} = \frac{\pi^2}{3}$$

10. **Ans (A)**

To find $\int_0^{\frac{\pi}{2}}\pi^2\frac{x^2}{\sin^2 x}$

$$= \pi^2\int_0^{\frac{\pi}{2}}x^2\cdot\text{cosec}^2 x$$

$$= \pi^2\left[\left(x^2(-\cot x)\right)_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}}2x\cdot\cot x\ dx\right]$$

$$= \pi^2\left[0+(2x\cdot\ln\sin x)_0^{\pi/2} - \int_0^{\frac{\pi}{2}}2\cdot\ln\sin x\ dx\right]$$

$$= \pi^2\left[0-2\left(-\frac{\pi}{2}\ln 2\right)\right] = \pi^3\ln 2$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (1.50)**

$$f(x) = x^2 - \int_0^1 (f(t)+x)^2\ dt$$

$$f(x) = x^2 - \int_0^1 (f^2(t)+x^2+2xf(t))\,dt$$

$$= x^2 - \int_0^1 f^2(t)dt - x^2 - 2x\int_0^1 f(t)\,dt$$

$$f(x) = -\int_0^1 f^2(t)dt - 2x\int_0^1 f(t)dt$$

Let $a = \int_0^1 f^2(t)dt$ and $b = \int_0^1 f(t)dt$

$\Rightarrow f(x) = -a - 2bx$

$$a = \int_0^1 (-a-2bt)^2 dt$$

$$a = a^2 + \frac{4b^2}{3} + \frac{4ab}{2} \quad .....(1)$$

$$b = \int_0^1 (-a-2bt)dt$$

$$b = -a - b$$

$$2b = -a \quad .....(2)$$

$$a = a^2 + \frac{1}{3}a^2 + 2a\left(-\frac{a}{2}\right)$$

$$\Rightarrow a = 0 \quad \text{or } a = 3$$

$a = 0$ is not possible so $a = 3$, $b = -\frac{3}{2}$

$$\Rightarrow f(x) = -3 + 3x = 3(x-1)$$

$$a = \int_0^{\infty} \frac{\tan^{-1}x}{x(1+x^2)}dx$$

$$x = \tan\theta \; a = \int_0^{\pi/2} \frac{\theta}{\tan\theta}d\theta$$

$$= [\theta \ell n \sin\theta]_0^{\pi/2} - \int_0^{\pi/2} \ell n \sin\theta d\theta$$

$$a = \frac{\pi}{2}\ell n 2$$

$$b = \int_0^1 \frac{5x^8 + 3x^4 + 1}{\sqrt{x^8 + x^4 + 1}}dx$$

$$= \int_0^1 \frac{5x^9 + 3x^5 + x}{\sqrt{x^{10} + x^6 + x^2}}dx$$

$$x^{10} + x^6 + x^2 = t$$

$$(10x^9 + 6x^5 + 2x)\,dx = dt$$

$$\left(5x^9 + 3x^5 + x\right)dx = \frac{dt}{2}$$

$$b = \int_0^3 \frac{dt}{2\sqrt{t}} = \frac{1}{2}\frac{t^{\frac{1}{2}}}{\frac{1}{2}}\Bigg|_0^3 = \sqrt{3}$$

For Question 11

$$\int_1^3 3(x-1)dx = 3\left(\frac{x^2}{2} - x\right)\Bigg|_1^3$$

$$= 3\left(\frac{9}{2} - 3 - \frac{1}{2} + 1\right) = 3(4-2) = 6$$

2. **Ans (4.50)**

$$c = \int_0^2 \sqrt{x^3+1}\;dx + \int_1^3 (x^2-1)^{\frac{1}{3}}dx$$

$$f(x) = (x^3+1)^{\frac{1}{2}},\; f^{-1}(x) = (x^2-1)^{\frac{1}{3}}$$

$$c = 2f(2) - 0f(0) = 2(9)^{\frac{1}{2}} = 6$$

$$d = \int_0^1 \frac{3x^2 + 9x + 7}{(x^2+3x+2)^3}$$

$$= \int_0^1 \frac{3x^2 + 9x + 7}{(x+2)^3(x+1)^3}dx$$

$$= \int_0^1 \frac{1}{(x+1)^3} - \frac{1}{(x+2)^3}dx$$

$$= \frac{1}{2}\left(\frac{1}{(x+2)^2} - \frac{1}{(x+1)^2}\right)\Bigg|_0^1$$

$$d = \frac{11}{36}$$

For Question 12

$$y = 3x - 3$$

$$x = \frac{y+3}{3} \quad \Rightarrow f^{-1}(x) = \frac{x+3}{3} = \frac{x}{3} + 1$$

$$A = \int_2^6 \left(\frac{x}{3} + 1\right)dx = \left(\frac{x^2}{6} + x\right)\Bigg|_2^6$$

$$= \frac{28}{3}$$

3. **Ans (2.00)**

$$\int \frac{dx}{\sqrt{a^{5x}}\sqrt[4]{(a^{2x}+a^{-2x})^3}}$$

$$\left(\text{where } t=\sqrt{2+\sin^{1/3}x}\right)$$

$$=\frac{1}{\ln a}\int \frac{dt}{t\cdot t^{5/2}\{t^2+t^{-2}\}^{3/4}}$$

$$\Rightarrow a^x=t \Rightarrow a^x \ln a\cdot dx=dt$$

$$=\frac{1}{\ln a}\int \frac{dt}{t^{7/2}\frac{(1+t^4)^{3/4}}{t^{3/2}}}$$

$$\left(\because \frac{m+1}{n}-\frac{p}{q}-\text{intger}\ \frac{-2+1}{4}-\frac{3}{4}=-1\right)$$

$$=\frac{1}{\ln a}\int \frac{dt}{t^2(1+t^4)^{3/4}}$$

$$\frac{1}{t^4}=1=u^4 \Rightarrow \frac{1}{t^5}dt=4u^3\ du$$

$$=\frac{1}{\ln a}\int \frac{-u^3\ du}{u^3}$$

$$=-\frac{1}{\ln a}\left\{\frac{1}{t^4}+1\right\}^{\frac{1}{4}}+C$$

$$=\frac{1}{\ln a}\{a^{-4x}+1\}^{\frac{1}{4}}+C$$

4. **Ans (8.00)**

$$I=\int \sqrt{\text{cosec}\,x}\left(2\sqrt[3]{\text{cosec}\,x}+1\right)^{\frac{3}{2}}\cdot\cos x\,dx$$

$$=\int \frac{(2+\sqrt[3]{\sin x})^{\frac{3}{2}}}{\sin^{1/3}x}\cdot\frac{\cos x}{\sin^{2/3}x}dx$$

Let

$$2+\sin^{\frac{1}{3}}x=t^2$$

$$\frac{1}{3}\frac{\cos x}{\sin^{\frac{2}{3}}x}dx=2t\ dt$$

$$I=\int \frac{t^3\cdot 6t}{t^2-2}dt=6\int\frac{t^4-4+4}{t^2-2}dt$$

$$=6\int\left[t^2+2+\frac{4}{t^2-2}\right]dt$$

$$=6\left[\frac{t^3}{3}+2t+\frac{4}{2\sqrt{2}}\ln\left|\frac{t-\sqrt{2}}{t+\sqrt{2}}\right|\right]+C$$

$$=2t^3+12t+6\sqrt{2}\ln\left|\frac{t-\sqrt{2}}{t+\sqrt{2}}\right|+C$$

where $t=\sqrt{2+\sin^{\frac{1}{3}}x}$

5. **Ans (261.00)**

Let $f(x)=ax^3+bx^2+cx+d$

It is passing through origin then $d=0$

$$f'(x)=3ax^2+2bx+c$$

$$f''(x)=6ax+2b$$

$$f''(0)=0+2b=0 \Rightarrow b=0$$

$$f'\left(\frac{-1}{\sqrt{10}}\right)=0$$

$$\Rightarrow 3a\left(\frac{-1}{\sqrt{10}}\right)^2+0+c=0$$

$3a+10c=0$ ....(i)

$f(1)=7 \Rightarrow a+c=7$ ....(ii)

from (i) and (ii)

$a=10$ and $c=-3$

$f(x)=10x^3-3x$

$$f(3)=10\cdot 3^3-3\times 3=270-9=261$$

6. **Ans (14.00)**

$$\frac{dy}{dx}+f'(x)\cdot y=(f(x)+1)f'(x)$$

$$IF=e^{\int f'(x)\,dx}=e^{f(x)}$$

$$y\cdot e^{f(x)}=\int e^{f(x)}\ (f(x)+1)f'(x)\ dx$$

Let $f(x)=t$

$$f'(x)dx=dt$$

$$y\cdot e^t=\int e^t\ (t+1)\,dt$$

$$y\cdot e^t=t\cdot e^t+C$$

$$g(x)e^{f(x)}=f(x)e^{f(x)}+C$$

$$g(x)e^{10x^3-3x}=(10x^3-3x)\,e^{(10x^3-3x)}+C$$

Put $x=0$

$$1\cdot e^0=0+c \Rightarrow c=1$$

$$g(1)e^7=7\cdot e^7+1$$

$$g(1)=7+e^{-7}=\alpha+e^{-\beta}$$

$$\Rightarrow \alpha+\beta=14$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (12)

Circle $x^2 + y^2 - 4x + 2y + 1 = 0$

or $(x - 2)^2 + (y + 1)^2 = 4$

So $x \in [0, 4]$

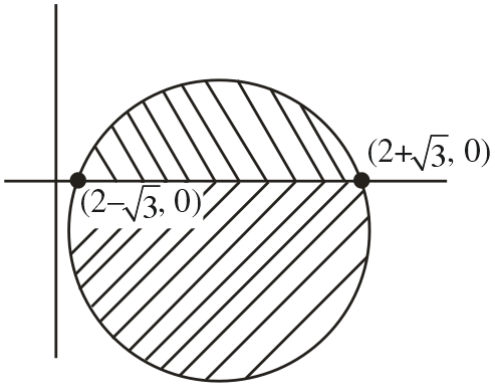
$$0 \leqslant \frac{x^3}{101} + \frac{x}{36} < 1, \quad \forall x \in [0, 4]$$

$$\left[\frac{x^3}{101} + \frac{x}{36}\right] = 0 \Rightarrow y = 0$$

Now at x-axis $y = 0$

$(x - 2)^2 = 3$

it cuts the x-axis at points $(2 - \sqrt{3}, 0)$ and $(2 + \sqrt{3}, 0)$



Let

$$A = \int_{2-\sqrt{3}}^{2+\sqrt{3}} \left(\sqrt{4 - (x-2)^2} - 1\right) dx$$

$$= \frac{4\pi - 3\sqrt{3}}{3}$$

The required ratio

$$= \frac{A}{4\pi - A} = \frac{4\pi - 3\sqrt{3}}{8\pi + 3\sqrt{3}}$$

8. Ans (2)

$$x\frac{dy}{dx} + y = \frac{2\cos x + 3}{(2 + 3\cos x)^2}$$

$$\frac{dy}{dx} + \frac{1}{x}y = \frac{2\cos x + 3}{x(2 + 3\cos x)^2}$$

$$IF = e^{\int \frac{1}{x} dx} = e^{\ln x} = x$$

$$y \cdot x = \int x \cdot \frac{\cos x + 3}{x(2 + 3\cos x)^2} dx$$

$$y \cdot x = \frac{\sin x}{2 + 3\cos x} + C$$

$$f\left(\frac{\pi}{2}\right) = \frac{2}{\pi}$$

$$\frac{2}{\pi} \cdot \frac{\pi}{2} = \frac{1}{2} + C \Rightarrow C = \frac{1}{2}$$

$$f(x) = \frac{1}{x}\left[\frac{\sin x}{2 + 3\cos x}\right] + \frac{1}{2x}$$

$$\pi \cdot f\left(\frac{\pi}{3}\right) = \left[\frac{\sqrt{3}}{2\left(2 + \frac{3}{2}\right)} + \frac{1}{2}\right] 3$$

$$\pi \cdot f\left(\frac{\pi}{3}\right) = \left(\frac{2\sqrt{3} + 7}{14}\right) 3 = 2.24$$

$[2.24] = 2$ Ans.

9. Ans (1)

Let $g(x) = \int_0^x f(t)\ dt \Rightarrow g'(x) = f(x)$

$$\lim_{x\to\infty} g(x) = \lim_{x\to\infty} \frac{e^x g(x)}{e^x}$$

$$= \lim_{x\to\infty} \frac{e^x (g + g')}{e^x} = \lim_{x\to\infty} (g + g') = L$$

$$\because \lim_{x\to\infty} g(x) = \lim_{x\to\infty} (g + g' - g')$$

$$L = \lim_{x\to\infty} (g + g') - \lim_{x\to\infty} g'$$

$$\Rightarrow \lim_{x\to\infty} g'(x) = 0$$

$$\Rightarrow \lim_{x\to\infty} f(x) = 0$$